

Culverts/Orifices

The equation used for culvert/orifice structures is:

$$Q = C_o A \sqrt{\frac{2gh}{k}} \times Nb$$

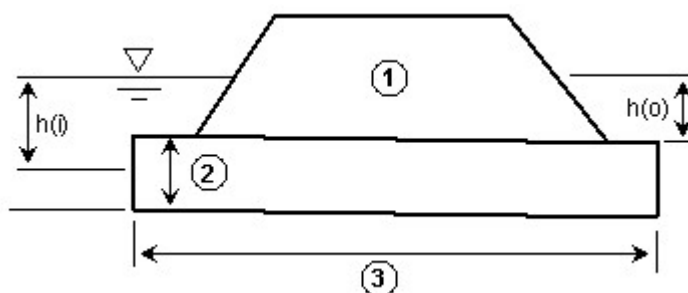
Where:

Under inlet control	Under outlet control
Q = Discharge (cfs)	Q = Discharge (cfs)
A = Culvert area (sqft)	A = Culvert area (sqft)
h = Distance between the water surface and the centroid of the culvert barrel (1/2 flow depth during partial flow) (ft)	h = Distance between the upstream and downstream water surface
Nb = Number of barrels	Nb = Number of barrels
Co = Orifice coefficient	Co = 1
k = 1	k = 1.5 + [(29n ² L)/R ^{1.33}] Where: n = Manning's n-value L = Culvert length (ft) R = Area/wetted perimeter (ft)

NOTE:

When a non-zero tailwater (TW) elevation is entered, Hydraflow Hydrographs Extension compares the pond stage with TW and computes a tailwater head, h_{TW}. If this head is less than the head computed as h, then h = h_{TW}.

The following illustration shows a profile of a typical culvert, where h(i) is the head under inlet control and h(o) is the head under outlet control.



(1) Embankment	(3) Pipe length	$h(o)$ - Head under outlet control
(2) Rise	$h(i)$ - Head under inlet control	

During the calculation process, both inlet and outlet control are evaluated. Inlet control means that the inlet of the culvert controls the amount of flow the culvert can handle. Under inlet control, the discharge depends on the barrel shape, cross-sectional area and inlet edge. Outlet control means that flow can enter the structure at a faster rate than it can exit. Under outlet control, the discharge depends on the slope, length and roughness of the barrel.

Hydraflow Hydrographs Extension computes the discharge at each stage, including intermediate stage points that it generates, using both inlet and outlet control equation parameters. The smallest value is used as the discharge at that elevation. This is reflected on the screen tabulation as "ic", inlet control and "oc", outlet control.

NOTE:

Hydraflow Hydrographs Extension does not assume full flow when the depth is actually partial.